

Data Preprocessing and Potential Changes Detection in Soil Surface Roughness of Crop Fields with Synthetic Aperture Radar (SAR)

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1. Abstract

Soil surface roughness is one of the main factors affecting the reflection of Synthetic Aperture Radar (SAR) backscatter. The ability to detect changes in roughness benefits data users in several ways. Some soil moisture models assume a constant in soil roughness over a specific interval. Knowing when the change occurs could avoid violating the models' assumption. Also, monitoring the changes can help us track farming seasons as plowing changes the surface. In this project, we find potential occurrences of the changes by utilizing SAR backscatter data from the Sentinel-1. First, we pick samples of crop fields using QGIS, a geographic information system software. Then, we retrieve Sentinel-1 backscatter data from Google Earth Engine and calculate the average of the signal over each field and the period-over-period change. Finally, we use outlier detection and clustering methods to analyze and find periods and fields where a roughness change might occur. For evaluation, we used a farm activity log of a farmer in the Netherlands to see how well the algorithm can detect changes caused by the activities.

2. Problem statement

Synthetic Aperture Radar (SAR) works by emitting microwave signals to the earth's surface and measuring the reflection called backscatter. Surface roughness strongly influences the signal, and sudden changes in roughness can brighten or darken the backscatter. If these changes are not appropriately modeled, they can result in false attribution of the effect, such as a wet or dry bias in soil moisture estimation. Because of this, this project aims to detect the period when the surface might change its roughness. Areas prone to change are architectural plots as farmers prepare their lands for a new cropping season. The main obstacle is that surface roughness is not widely studied, and datasets are unavailable. Gathering the data ourselves is also impossible under the project timeline and resources as we need to travel to multiple fields and repeat measuring surface parameters over a long period. Thus, we focused on preprocessing the satellite data and employed unsupervised methods to analyze the data.

3. Research Questions

To what extent can we rely on outlier detection methods to distinguish soil surface roughness changes due to farming activities among the changes in period-over-period backscatter?

4. Methodology

This project follows the method described in the CRISP-DM framework.

4.1 Business Understanding

Synthetic-aperture radar (SAR) is an active sensor that emits microwave signals and records the reflection from the observing surface. Unlike real-aperture radar, synthetic radar uses satellite motion to enable the sensor to produce high-resolution images as if it has a large antenna. The SAR operates by capturing the signal in a side-looking direction.

SAR records both the amplitude and phase of the reflected wave. In the case of amplitude, the sensor registers the wave energy. The reflected signal could be low if the emitted signal reflects away when the surface is smooth. On the other hand, if it gets reflected twice when it hits a corner between roads and buildings, the returned signal can be high. But most of the time, the signal gets attenuated when the surface is rough or when the signal gets absorbed in the soil or vegetation. For phase, the sensor data can be used for calculating the difference between two observations of the same position between two satellite visits using interferometric coherence (InSAR), indicating a change in surface level. These raw SAR data containing both amplitude and phase are called single look complex (SLC). The analysis of SLC data is more complex due to noise, called speckle, and large file size. An alternative product called Ground Range Detected (GRD) is simpler to use. Conceptually, it averages signals from multiple scans of the same position. The result is a reduction in noise, but the phase information and the resolution are reduced.

Agriculture lands can experience changes in roughness throughout their utilization. Farm activities such as tillage, planting, and harvest disturb the soil and may cause irregular changes in roughness. SAR might detect such a change over two consecutive satellite visits differently from the nearby area. The result might enable us to track land usage remotely over the years.

Still, there are remaining obstacles. Farm activities in the same region often occur in the same period, e.g., spring and summer. We may not be able to detect individual activity by identifying outliers.

In this project, the method will be applied to farming activities in Harrysfarm from 2017 to 2023. A work by Huber et al. 2023 [3] used a technique that utilized interferometric coherence as an indicator during 2017-2020 in the same farm and achieved an F-Score of 0.55 (note that they did not split the data into train-test sets). Because our method is easily scalable, our success criteria are to achieve a similar level of F-Score in the test set.

4.2 Data Understanding

In this project, we source different data to provide comprehensive information. We pulled most of the data from Google Earth's engine. Employing the service enables us to utilize the data without needing to preprocess big and complex satellites data.

4.2.1. Sentinel-1 SAR Backscatter Data

Sentinel-1 is a European satellite constellation in the Copernicus Programme. There were two satellites in orbit, Sentinel-1A and Sentinel-1B. Each satellite has a twelve-day revisit, thus six days when working together. However, Sentinel-1B experienced an issue with data transmission and retired in early 2022. The satellites are equipped with C-band SAR instruments with a wavelength of 5.5 cm.

Google Earth Engine offers Sentinel-1 derived data called Ground Range Detected (GRD). GRD contains the amplitude of the backscatter values which have been average over multiple looks, resulting in lower noise.

Moreover, Sentinel-1 also utilizes polarization. The transmitter and receiver can operate in different polarized orientations, horizontal (parallel to the surface) or vertical. Different polarization combinations are sensitive to specific reflection types. Conventionally, we denote the type of polarization with two letters, V and H. V refers to the vertical component of the signal while H refers to the horizontal. The preceding letter indicates the type of transmitting signal, and the succeeding letter refers to the receiving signal. For example, VH means transmitting a vertical signal while listening for only the horizontal component of the return signal. There are four possible combinations; VV, VH, HV, and HH. The availability of these polarizations depends on the specification of the satellite and also the area it surveys. For our area of interest (the Netherlands), Sentinel-1 data from Google Earth Engine contains only GRD data with only VV and VH polarization.

4.2.2. Sentinel-2 Normalized difference vegetation index

Sentinel-2 is also a European satellite constellation in the Copernicus Programme. Unlike Sentinel-1, these satellites use multispectral passive sensors. The sensors can capture 13 spectral bands, including visible light and near-infrared, in a 5-day revisit interval.

The normalized difference vegetation index (NDVI) is a metric that helps quantify vegetation health and density. We can calculate NDVI using the following definition,

$$NDVI = \frac{NIR - Red}{NIR + Red} \quad (1)$$

where NIR is the near-infrared spectral reflectance and Red is the visible red. A denser and greener canopy produces higher NDVI, while it is about 0.1 for bare soil.

4.2.3. Global Precipitation Measurement (GPM)

Global Precipitation Measurement (GPM) is a joint program between NASA (USA) and JAXA (Japan). The mission is to use satellites to monitor rainfall and snowfall around the globe within a frequency of every 30 minutes. The important measurements of GPM are the precipitation and the probability of liquid precipitation.

4.2.4. Google Earth Satellite Images

Even though Sentinel-2 can capture high-resolution images, at best 10 square meters per pixel, the resolution still needs to be higher to produce good boundaries of agriculture fields. Google Earth Satellite Images release much higher resolution, which could be under 1 square meter per pixel. However, depending on the area, the images may be dated and not illustrate the current state of the area.

4.2.5. Harrys Farm Activity Log

Harrysfarm [4] is a 50-hectare farm in Swifterbant, Netherlands. The farm owner, Harry Schreuder, has published a website where he has kept his farm activity log since 2002. He planted crops such as potato, onion, wheat, and sugar beet in rotation in five fields. Each year, the farmer marked a date, his activity, and the growth of the crop in each field. Activities like tillage, ridge, planting, and harvesting phase change surface roughness. The activity log was used in Huber et al. 2023 [3] for evaluation.

5. Data Preparation

5.1. Field Boundary

We use QGIS, a geographic information system software, to annotate sample field boundaries as polygons using images from Google Maps. We annotate 196 field boundaries. Then, we export the annotation as a shape file. We assume that the boundaries don't change during the study period.



Figure 1: Annotated farm boundaries located between Swifterbant and Lelystad, Netherlands. The location of Harrysfarm is highlighted with solid yellow.

5.2. Satellite data

We fetched the satellite data (Sentinel-1 SAR Data, Sentinel-2 NDVI, and GPM) from the Google Earth Engine with the following procedure:

- 5.2.1 For each field, we queried the time series data from 2017 to 2022 at the highest resolution for each pixel. Although the data are available until 2024, We pulled the data until only 2022 as both Sentinel-1A and Sentinel-1B were operating.
- 5.2.2 Pixels within the same field boundary are grouped. To smooth them out, we averaged the time series of all these pixels in the boundary together.
- 5.2.3 For Sentinel-1 SAR, VV and VH from Google Earth Engine are given in both decimal and decibel (dB). We calculated the change of both polarization channels, VV and VH, using difference and ratio as model features. Because various factors such as incidence angle, the shape of the surface, and alignment could influence the SAR measurement, we can perform the computation separately for ascending and descending orbits to reduce part of the difference. Also, we can use Incidence Angle Normalization (Bauer-Marschallinger et al. [2021]) to estimate the backscatter at a reference incidence angle of 38° . Incidence Angle Normalization is given as the following linear relationship.

$$\sigma^0(38, t) = \sigma^0(\theta, t) - \beta(\theta - 38) \quad (2)$$

Where σ^0 is the backscatter coefficient. $\beta = -0.13 \text{ dB}$ is used. θ is the incidence angle of the measurement.

5.2.4 Sentinel-2 NDVI and GPM data were used to provide additional context in the visualization. High NDVI values indicated dense vegetation coverage, while elevated GPM measurements corresponded to increased moisture content from both rainfall and snow. When snow covers the soil, the surface becomes highly reflective, potentially destabilizing the relationship between backscatter signals and soil moisture content. Consequently, our approach may not be suitable for snow-covered periods, making their exclusion reasonable. However, GPM measurements alone cannot reliably indicate snow cover.

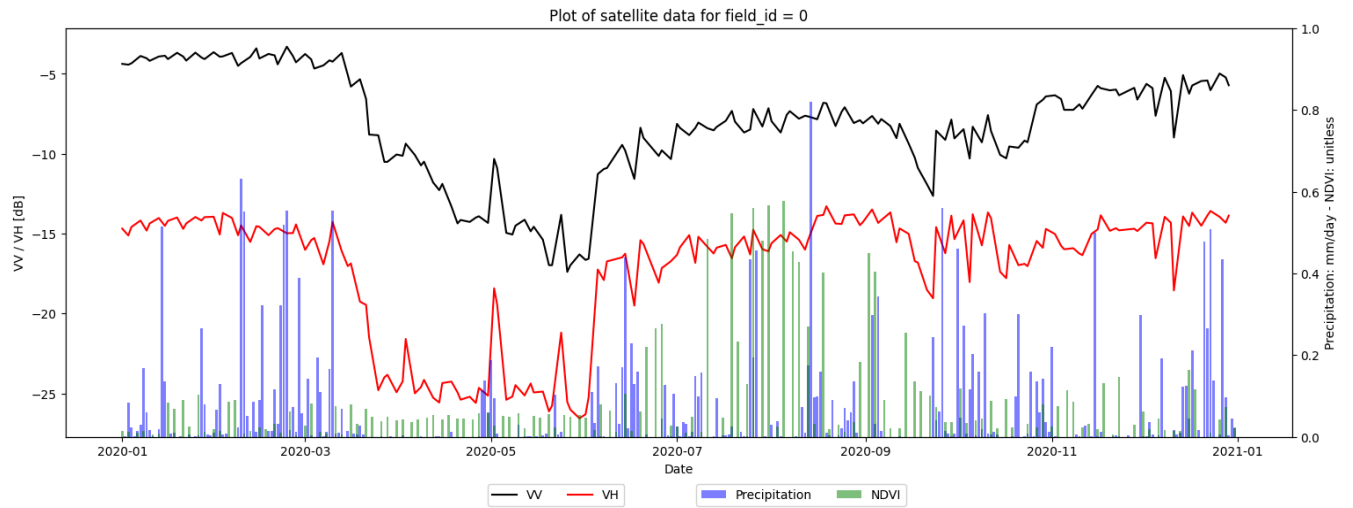


Figure 2: Sample data from Field ID = 0 in the year 2020.

Note: VV and VH are normalized using the Incidence Angle Normalization.

VH is usually low for rough surface scattering (Meyer 2019 [5]).

5.3. Harrysfarm's Farm Activity

The farmer logged his farm activity for each crop separately. For example, in 2017, potatoes were planted on 11 April in parcel 5 (denoted as 4 in figure 2). He then added entries for growth and further activities until the field was harvested in the respective year. In other years, the farmer practiced crop rotation so the parcel may be used to grow different crops, such as Winter Wheat in 2018 and sugar beet in 2019.

We collected five unique groups of activities from his logs from 2018 to 2022, including fertilize, harvest, plant, ridge, and tillage. We noted only the first day if any activities spanned over multiple days. There were 102 activities in total between 2018 and 2022.



Figure 3: Structure of Harrysfarm crop fields.

6. Modeling

6.1. Concept

Agricultural plots in the same neighborhood experiencing identical weather conditions should show similar changes in backscatter signals. For instance, when it rains, the soil becomes wetter, resulting in higher backscatter across all plots. However, when farmers prepare their land or plant crops, the soil surface of the affected areas changes differently from unaffected plots. Using this principle, we can detect changes in surface roughness by identifying plots with outlying period-over-period changes in backscatter signals, assuming farming activities occur at different times.

We used backscatter data from 196 fields shown in Figure 1 to provide context for weather and other natural changes affecting the backscatter. We created features for changes in backscatter between consecutive satellite visits, formed a distribution of the change in each period, and applied outlier detection models to the distribution to check if any Harrysfarm parcels were flagged as outliers. We marked the classification as true positive if any farming activity occurred at the parcels during the period.

6.2. Models

We applied three different multivariate outlier detection methods from the Python Scikit-learn machine-learning library.

2.1 Mahalanobis Distance with χ^2 Distributions

Mahalanobis Distance, $d(x)$, is a distance matrix that considers the interaction of each dimension of the data i.e. the shape of the overall distribution of the data.

It is defined as

$$d(x) = \sqrt{(x - \bar{x})^T S^{-1} (x - \bar{x})} \quad (3)$$

Where x is a new sample we want to find the distance. $\bar{x}$ is the data mean and S^{-1} is the data covariance matrix.

If we assume that the data follow a multivariate normal distribution, the Mahalanobis distance will follow χ^2 distribution with a degree of freedom equal to the number of the dimensions of the data.

Note that we should use a robust method to calculate the mean and the variance. For this, we will use the Minimum Covariance Determinant (MCD) method.

2.2 DBSCAN Clustering

Density-based spatial clustering of applications with noise (DBSCAN) is a data clustering method that attempts to find high-density regions surrounded by low-density gaps. It will recognize the high-density parts as clusters. We can consider samples outside the clusters as outliers.

2.3 Isolation Forest

Isolation Forest is a method in which we attempt to fit a random forest model over the data and determine how easy it is to split out a sample. To split the data, the forest randomly selects a feature and a split value. Then, it determines the average depth of the trees needed to isolate a sample. If the depth is shallow, i.e., if it is easy, then it might be an outlier.

6.3. Experiment Setup

For each model, we applied the model to each time point one by one using the feature calculated in section 6.5. Our hypothesis is that the crop fields in the same region should experience the same environmental influence, such as weather. However, if there is a farmer's activity, such as preparing or working on a field, the change in the soil surface should be different from that of other nearby land.

6.4. Data Split

We used data from 2017 - 2020 (31 events) as a training set and 2021 - 2022 (41 events) as a testing set.

6.5. Feature Set

Three feature sets suggested by Zhu et al. 2019 [1] are used. t and $t - 1$ denoted the dates on which the satellite data is used. If the feature is calculated with respect to orbit (ascending or descending), we calculate the difference and ratio with the lag observation of the same orbit. VV and VH values can be in decimals or decibels.

5.1 [VV Difference, VH Difference]

$$[VV_t - VV_{t-1}, VH_t - VH_{t-1}] \quad (4)$$

5.2 [VV Ratio, VH Ratio]

$$\left[\frac{VV_t}{VV_{t-1}}, \frac{VH_t}{VH_{t-1}} \right] \quad (5)$$

5.3 [VV Ratio, VH Ratio, VH/VV Ratio]

$$\left[\frac{VV_t}{VV_{t-1}}, \frac{VH_t}{VH_{t-1}}, \frac{VH_t}{VV_t} / \frac{VH_{t-1}}{VV_{t-1}} \right] \quad (6)$$

6.6. Hyperparameter Set

During the training, we used grid search to determine the best set of hyperparameters. The following lists were the range of values we used for the search. Hyper parameters are the parameters for Scikit Learn implementation of the outlier detection model.

6.1 DBSCAN

Epsilon (eps): [0.1, 0.3, 0.5, 0.7, 1.0]

Minimum points (min_samples): [3, 5, 10, 15, 20]

Leaf size of Nearest Neighbors algorithm (leaf_size): [10, 20, 30, 40, 50]

6.2 Isolation Forest

Number of base estimator (n_estimators): [30, 50, 100, 150]

Amount of contamination of the data set (contamination): [0.1, 0.2, 0.3, 0.4, 0.5, 'auto']

6.3 Mahalanobis Distance

Cutoff alpha value (Alpha): [0.01, 0.05, 0.1]

6.7. Experiment Procedure

We perform our experiment with the following setup.

For each feature set:

- a. For each preprocessing method:
 - i. For each outlier detection method:
 - For each period in the dataset:
 - a. Calculate the features for each crop field.
 - b. Apply standardization for DBSCAN as it is not scale invariant.
 - c. Apply the method to the features to get a list of outlier fields.
 - Check for periods when the fields in Harrysfarm (field ID 0 to 4) are marked as an outlier. The periods overlap as we calculate the features separately by ascending and descending orbits.
 - Evaluate the results by comparing the outlier periods with the annotated date from the Harrysfarm log. We count outlier periods containing any annotated date as a true positive. If there is more than one period containing a single farm event, we count them only once. All periods that cannot capture the annotated dates are false positives.
 - Calculate classification evaluation metrics for whether a detected outlier period contains an annotated farming activity.
 - b. Use grid search over the hyperparameter sets (from step 6.) to select the highest F-Score configuration for each method.

7. Result & Evaluation

7.1. Classification Metrics

We used classification evaluation metrics to evaluate the outlier methods.

7.1.1 Precision

Precision tells us about the concentrations of the surface roughness change among all the results received from the model. High precision ensures that the potential results are likely actual events without much detailed investigation. However, models with high precision but low recall could miss detecting most events.

$$Precision = \frac{TruePositives}{TruePositives + FalsePositives} \quad (7)$$

7.1.2 Recall

Recall tells us how many surface roughness events are discovered by outlier algorithms. A result with high recall allows researchers to filter candidates for potential changes in the roughness before using detailed and possibly more time-consuming analysis to pick the correct one. If recall is high but the precision is low, the models give diluted results, which may be useless.

$$Recall = \frac{TruePositives}{TruePositives + FalseNegatives} \quad (8)$$

7.1.3 F-Score

F-Score is a harmonic mean of precision and recall. F-Score can be used to rank the most balanced models.

$$F1 = 2 \cdot \frac{Precision \cdot Recall}{Precision + Recall} \quad (9)$$

7.2. Result

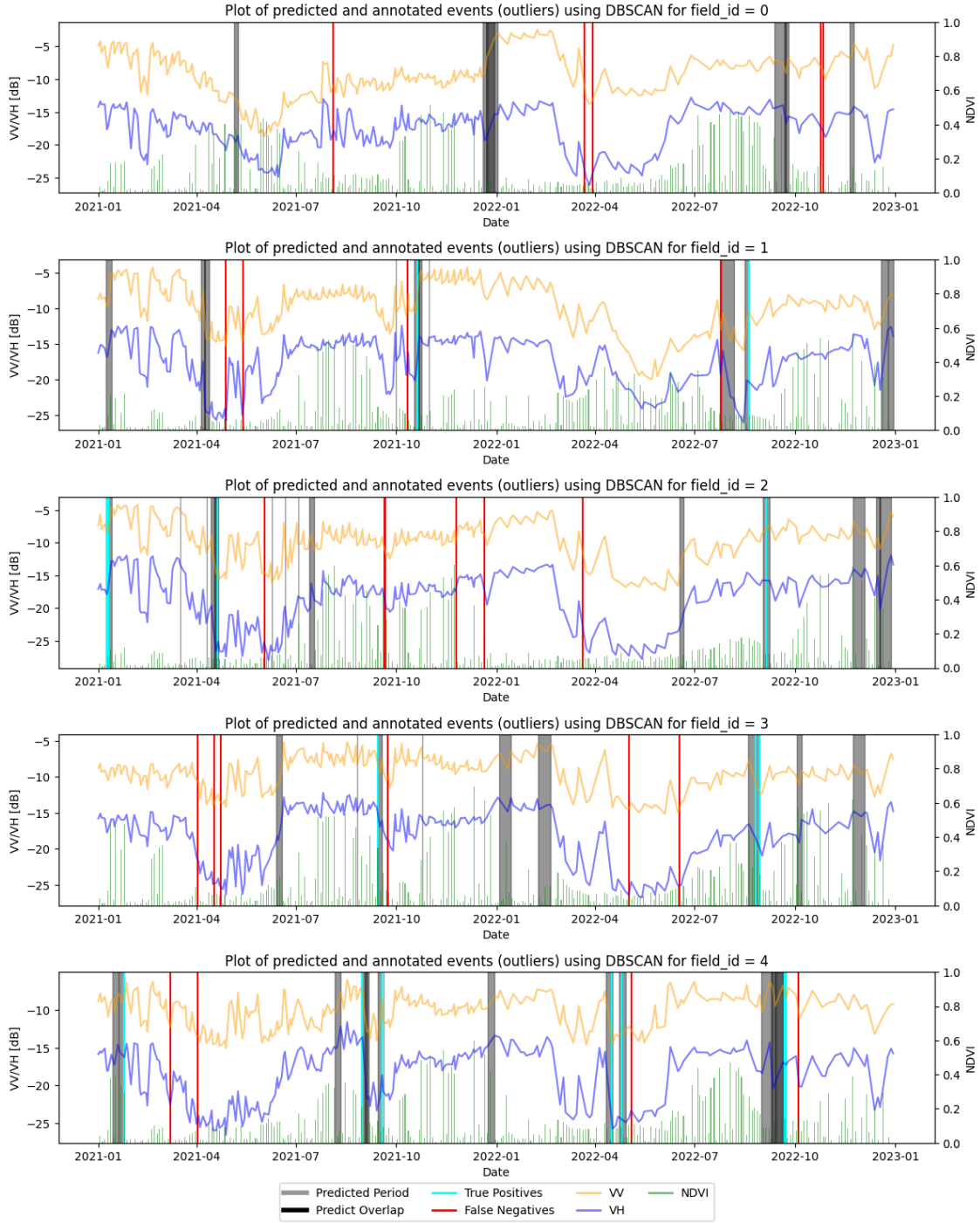


Figure 4: Testing set result using DBSCAN with F-Score=0.3265 from Table 5.
 Feature: [V Ratio, V Ratio] using decibel values and calculated by orbit.
 Incidence angles are normalized to 38°.

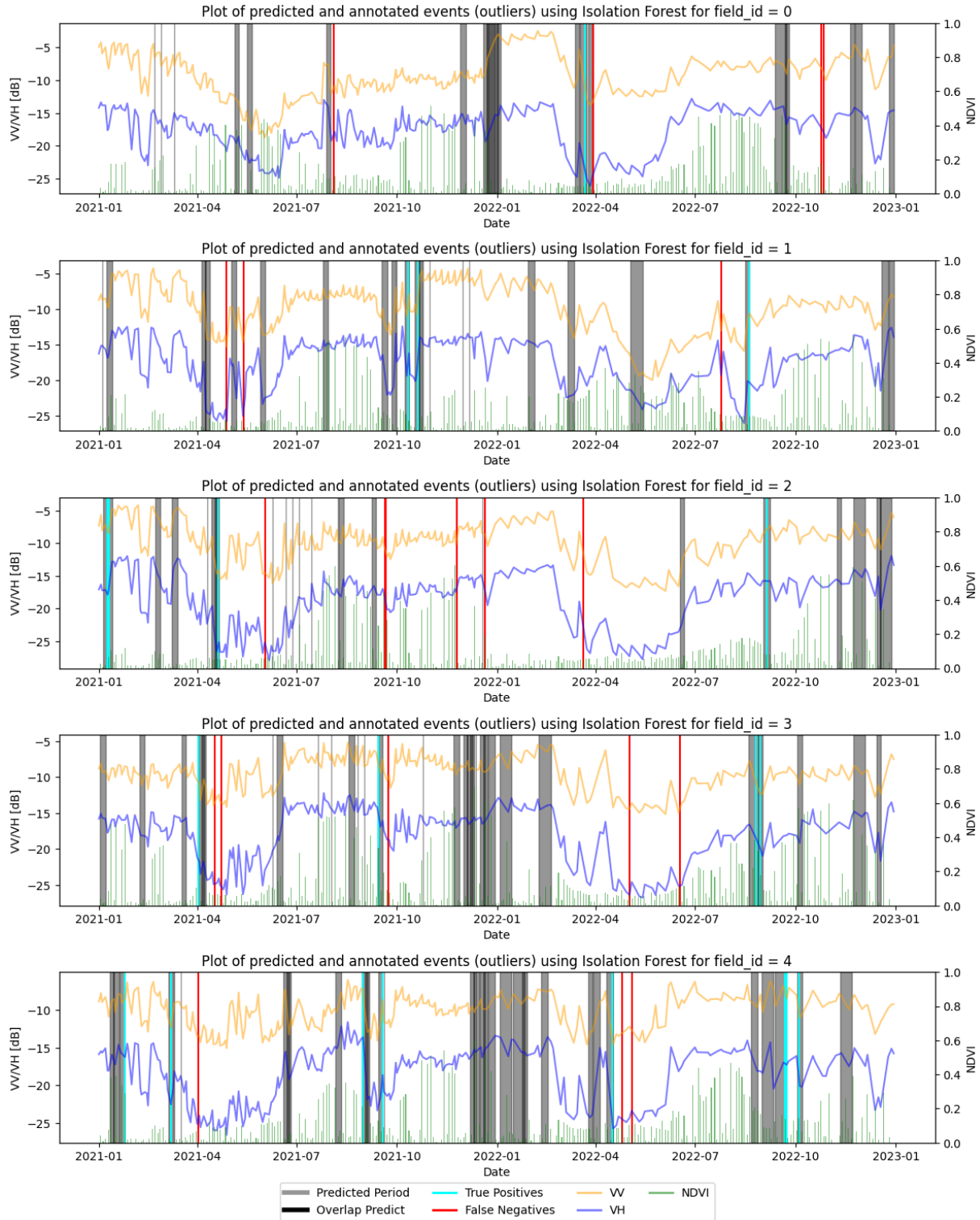


Figure 5: Testing set result using Isolation Forest with F-Score=0.2395 from Table 4.

Feature: [VV Ratio, VH Ratio] using decibel values and calculated by orbit.

Incidence angles are normalization to 38°.

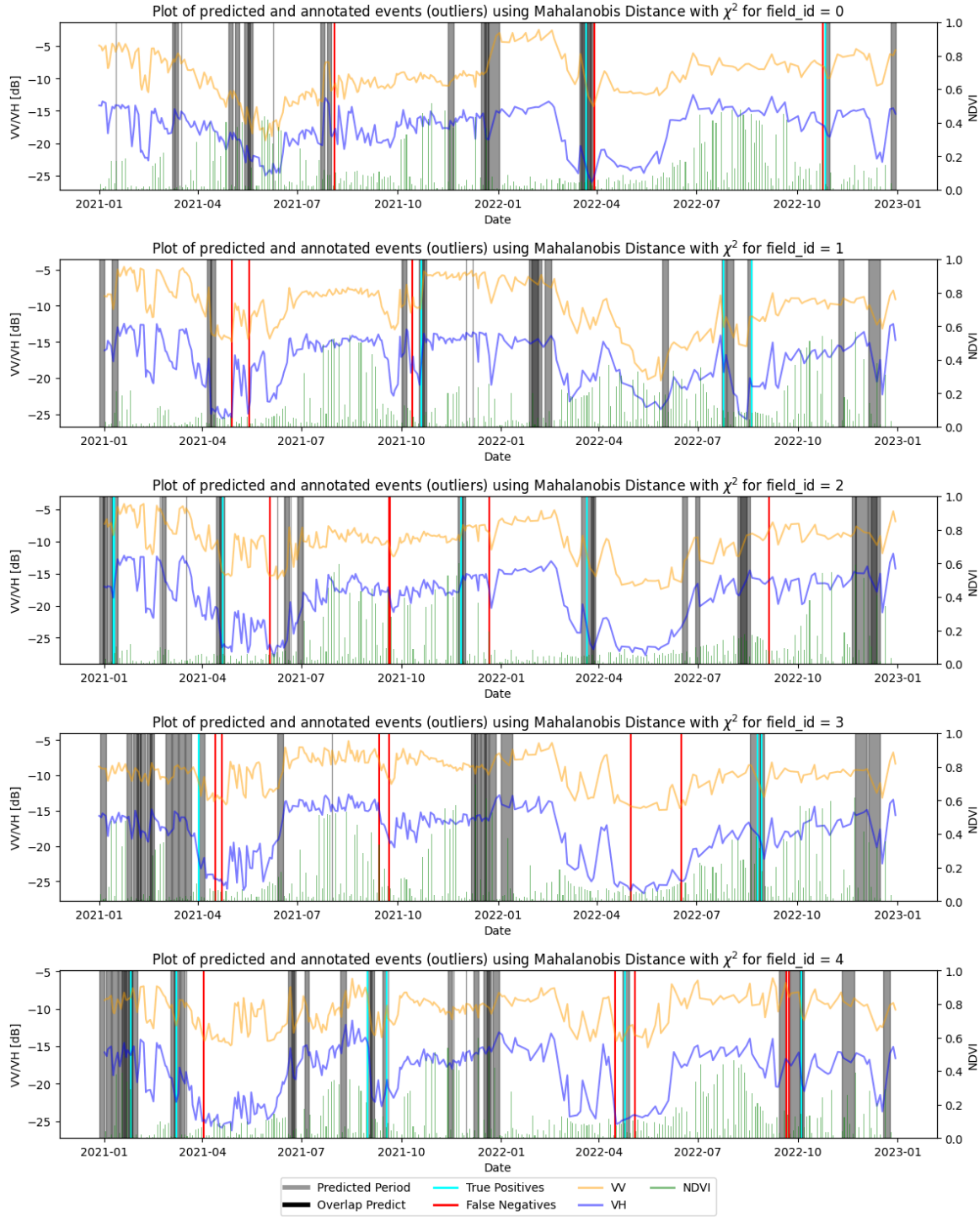


Figure 6: Testing set result using Mahalanobis Distance with χ^2 with F-Score=0.2346 from Table 4.
 Feature: [VV Difference, VH Difference] using decibel values and calculated by orbit.

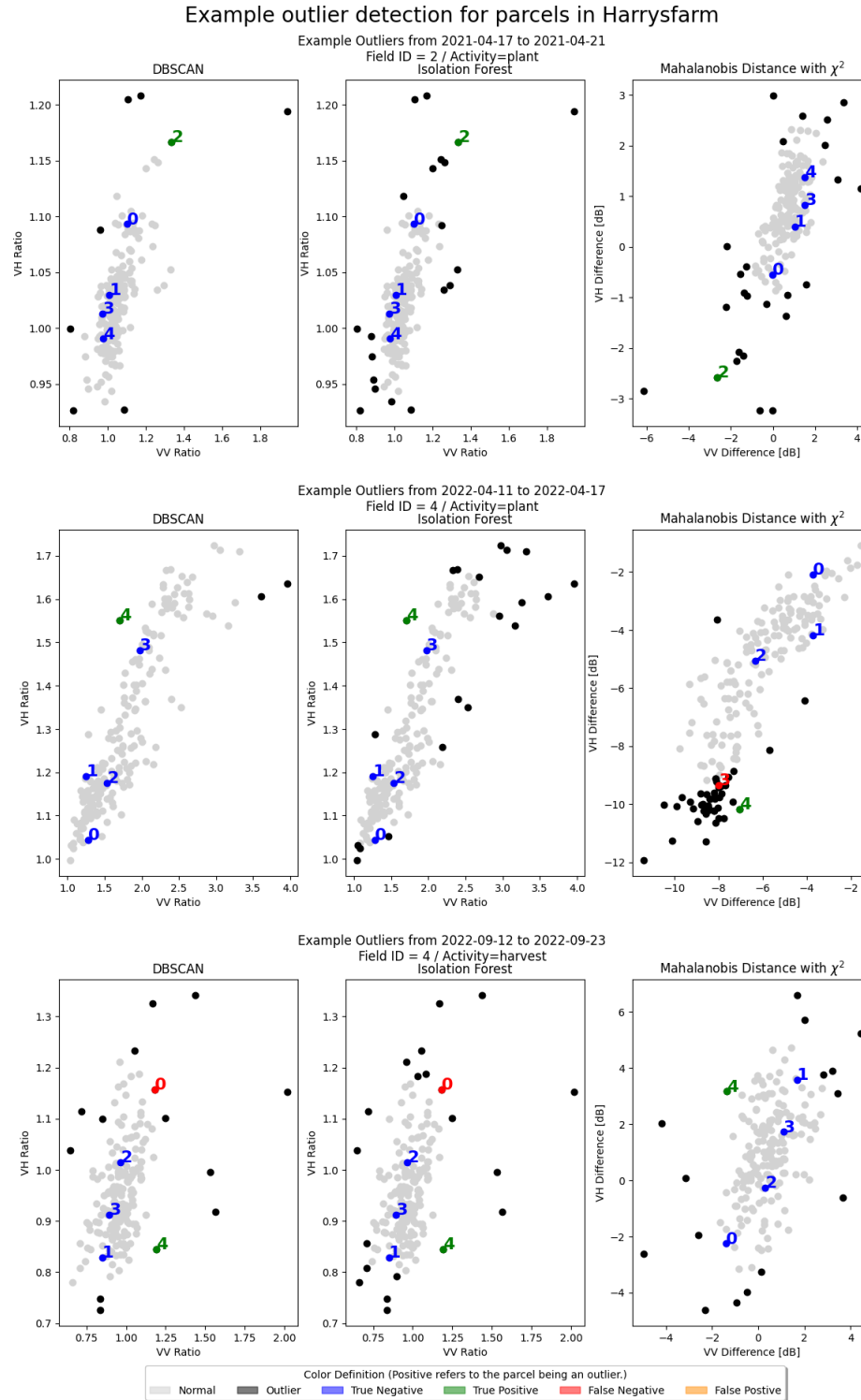


Figure 7: Example detected outliers using the parameter in Table 4.

Note that DBSCAN and Isolation Forest use [VV Ratio, VH Ratio] but Mahalanobis Distance with χ^2 uses [VV Difference, VH Difference].

Figures 4, 5, and 6 show the test set results for each outlier detection method with the best training-set F-Score from Tables 1, 2, and 3. These plots illustrate detection results from 2021 to 2023 with backscatter and NDVI for context. Each figure contains five plots per field. The plots display raw backscatter data in orange (VV polarization) and blue (VH polarization). Generally, VV backscatter is higher than VH in croplands since surface scattering dominates in sparse or low-growing vegetation, unlike forested areas where volume scattering is more significant. The green bars show NDVI values, which correlate with vegetation density. Tall bars (generally above 0.2) indicate grown crops, typically occurring in summer months. NDVI was included to show periods of bare soil versus vegetation cover. Grey bands indicate outlier detection results, which can overlap because outliers were calculated from differences between consecutive satellite visits in the same ascending/descending orbit direction. Farming activities appear as vertical lines - cyan for true positives (activities within grey bands) and red for false negatives.

Figure 7 demonstrates example results for three different activities through scatter plots showing data points in the feature spaces of each detector. Each dot represents a parcel, with grey points indicating normal data and black points showing outliers. Colored dots represent parcels in Harrysfarm: green for true positives (detected outliers), red for false negatives (undetected outliers), orange for false positives (detected outliers without farming activity), and blue for parcels without activity or detection. The figure illustrates differences between detectors and feature spaces (using a difference or a ratio of backscatter). For instance, DBSCAN typically detects fewer outliers than Isolation Forest in the same feature space. The distribution shapes also differ between the difference and ratio of backscatter.

7.2.1 Training Set Result

Model	Feature	Preprocessing	Precision	Recall	F-Score
DBSCAN	[VV Difference, VH Difference]	Decimal (by orbit)	0.1261	0.2459	0.1667
		Decibel	0.0614	0.2295	0.0969
		Decibel (38° Norm.)	0.0622	0.2131	0.0963
		Decibel (by orbit)	0.1478	0.2787	0.1932
		Decibel (by orbit & 38° Norm)	0.1466	0.2787	0.1921
	[VV Ratio, VH Ratio]	Decimal (by orbit)	0.1111	0.2787	0.1589
		Decibel	0.0719	0.1639	0.1000
		Decibel (38° Norm.)	0.0811	0.1475	0.1047
		Decibel (by orbit)	0.1650	0.2787	0.2073
		Decibel (by orbit & 38° Norm)	0.1569	0.3934	0.2243
	[VV Ratio,	Decimal	0.0946	0.2295	0.1340

	VH Ratio, VV/VH Ratio]	Decibel	0.0661	0.1311	0.0879
		Decibel (38° Norm.)	0.0656	0.1311	0.0874
		Decibel (by orbit)	0.1269	0.2787	0.1744
		Decibel (by orbit & 38° Norm)	0.1223	0.2787	0.1700

Table 1: Training Result for DBSCAN.
Bold numbers denote the best result.

Model	Feature	Preprocessing	Precision	Recall	F-Score
Isolation Forest	[VV Difference, VH Difference]	Decimal (by orbit)	0.0798	0.3443	0.1296
		Decibel	0.0720	0.3115	0.1169
		Decibel (38° Norm.)	0.0703	0.2951	0.1136
		Decibel (by orbit)	0.1131	0.5082	0.1851
		Decibel (by orbit & 38° Norm)	0.1124	0.4918	0.1829
	[VV Ratio, VH Ratio]	Decimal	0.0915	0.4262	0.1507
		Decibel	0.0642	0.2787	0.1043
		Decibel (38° Norm.)	0.0677	0.2787	0.1090
		Decibel (by orbit)	0.1142	0.4754	0.1841
		Decibel (by orbit & 38° Norm)	0.1208	0.4754	0.1927
	[VV Ratio, VH Ratio, VV/VH Ratio]	Raw measurement	0.0808	0.5246	0.1400
		Decibel	0.0654	0.2787	0.1059
		Decibel (38° Norm.)	0.0556	0.3607	0.0963
		Decibel (by orbit)	0.1042	0.4426	0.1687
		Decibel (by orbit & 38° Norm)	0.1070	0.4262	0.1711

Table 2: Training Result for Isolation Forest.
Bold numbers denote the best result.

Model	Feature	Preprocessing	Precision	Recall	F-Score
Mahalanobis Distance with χ^2	[VV Difference, VH Difference]	Decimal (by orbit)	0.0519	0.4590	0.0933
		Decibel	0.0607	0.3443	0.1032
		Decibel (38° Norm.)	0.0612	0.3443	0.1040
		Decibel (by orbit)	0.0927	0.6230	0.1614
		Decibel (orbit & 38° Norm)	0.0905	0.6066	0.1574
	[VV Ratio,	Decimal (by orbit)	n/a (Covariance matrix wasn't full rank)		

	VH Ratio]	Decibel	0.0525	0.3115	0.0898
		Decibel (38° Norm.)	0.0543	0.3279	0.0932
		Decibel (by orbit)	0.0878	0.6230	0.1538
		Decibel (by orbit & 38° Norm)	0.0864	0.6066	0.1513
	[VV Ratio, VH Ratio, VV/VH Ratio]	Decimal (by orbit)	0.0663	0.5574	0.1185
		Decibel	0.0510	0.3770	0.0898
		Decibel (38° Norm.)	0.0494	0.3607	0.0870
		Decibel (by orbit)	0.0747	0.6230	0.1333
		Decibel (by orbit & 38° Norm)	0.0737	0.6066	0.1314

Table 3: Training Result for Mahalanobis Distance with χ^2 .
Bold numbers denote the best result.

The training results clearly show that using the features in decibels by orbit direction is better. The best method was to use the ratio of backscatter for DBSCAN and Isolation Forest, while the difference of backscatter was better for Mahalanobis Distance with χ^2 . Note that only the training results from the best hyperparameters found from grid search for each input feature were shown in the table.

7.2.2 Testing Set Result

The configurations with the highest F-Score of each model from the training set were tested against the testing set.

Model	Feature	Preprocessing	Hyper Parameters	Test Set Result		
				Precision	Recall	F-Score
DBSCAN	[VV Ratio, VH Ratio]	Decibel (by orbit & 38° Norm)	eps=0.5 min_samples=3 metric="euclidean" leaf_size=10	0.2807	0.3902	0.3265
Isolation Forest	[VV Ratio, VH Ratio]	Decibel (by orbit & 38° Norm)	n_estimators=50 contamination=0.1	0.1587	0.4878	0.2395
Mahalanobis Distance with χ^2	[VV Difference, VH Difference]	Decibel (by orbit)	alpha=0.01	0.1570	0.4634	0.2346

Table 4: Test Result for the best configuration for each model.
Bold numbers denote the best result.

The DBSCAN performed the best in both the training set and the test set, with the best F-Score of 0.2243 and 0.3265, respectively. DBSCAN also produced the most balanced

result in both precision and recall. Isolation Forest came second with an F-Score of 0.1927 for the best training result and 0.2346 for the test set. The last place was from Mahalanobis distance with χ^2 , achieve the best at 0.1614 in the training set and 0.1538 in the test set. While the last two models had much higher recall than the DBSCAN, they got much lower precision at only about 0.15.

7.3. Discussion

We can compare our result with Huber et al. 2023 [3] in which they produced a recall of 0.6 and an F-Score of 0.55 during 2017-2022. The period was the same as our training set. We can see that our results appear modest compared to them. The best F-Score was from the DBSCAN at only 0.22. Although there were some configurations of Mahalanobis Distance with χ^2 method that produced a recall at 0.62, the F-Score was only 0.15. Thus, we did not successfully reach our goal. However, given current tools and data availability, our approach is more practical if employed on a larger scale. Google Earth Engine provides quick access to backscatter data and computation of soil surface roughness changes. In contrast, Huber et al. 2023 [3] used Single Look Complex (SLC) data, requiring manual downloads from Copernicus and specialized InSAR preprocessing software.

Limited annotated data constraints further method improvement and verification. Gathering more data through surveys or field observation would require significant time, resources, and funding. Additionally, concurrent farming activities in the region might invalidate our outlier-based assumption. The farming log may also miss some roughness change events, such as crop budding or frozen soil effects. More careful data collection and inspection are needed for verification.

Attempts to use contextual data to filter incompatible periods were unsuccessful. While we tried using NDVI and precipitation data to identify periods where SAR might produce an unstable value, the relationship between these factors and annotated events remained unclear. Figures 4, 5, and 6 show frequent false positives during winter months in 2022 and 2023, possibly due to frozen fields. GPM data was also explored for winter month filtering, but field freezing periods remained ambiguous.

8. Deployment

The source code of the project is available at https://github.com/gentlerainsky/soil_roughness_change_detection.

9. Domain-specific Lectures

Data Retrieval in Earth Observation (120.110) was taken and passed in the Summer Semester of 2023.

10. Reference

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